

Sami Ahmed Mohammed

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AI/ML Engineer with experience building end-to-end AI products across GenAI, NLP, computer vision, and predictive ML. Skilled in model development, cloud deployment, APIs, MLOps, and production AI workflows using Python, AWS, GCP, TensorFlow, PyTorch, and Hugging Face.

EXPERIENCE

University of Central Florida

Orlando, FL

Graduate Teaching Assistant

August 2026 - Present

- Support STA 2023 coursework and assessments, applying statistical reasoning to grade consistently and reinforce students' understanding of core methods.
- Guide students through quantitative problems during office hours, clarifying statistical concepts and improving confidence in interpreting data and results.
- Prepare lecture materials and course resources for Statistical Methods I, helping organize instruction around applied analysis, inference, and problem solving.

SEO Content AI

Los Angeles, CA

Machine Learning Engineer Intern

April 2023 - December 2023

- Built production AI workflows using GPT-3.5, FLAN-T5, BERT, RoBERTa, and DistilBERT, enabling automated content generation, refinement, and quality improvement.
- Developed NLP pipelines for keyword extraction, topic clustering, grammar correction, validation, and multilingual generation, streamlining end-to-end content operations.
- Integrated Python and JavaScript services with browser and web applications, turning model outputs into responsive user-facing AI features.
- Deployed containerized AI services on Docker, AWS ECS, and Fargate, enabling scalable inference and more reliable production execution.
- Optimized generation pipelines and service orchestration, improving content-processing speed by 40% while reducing latency across production AI workflows.
- Integrated multilingual generation and translation workflows, expanding product reach across international users while preserving consistent content-processing behavior.
- Partnered across model, backend, and product layers to convert AI capabilities into deployable features aligned with user and business requirements.

Dhaaps Edutech

Hyderabad, TG

Data Science Intern

May 2023 - June 2023

- Developed a loan-approval prediction solution, applying machine learning to improve approval accuracy by 20% and support more consistent decisions.
- Built Python-based preprocessing and analytics workflows, automating repetitive data preparation tasks and improving efficiency across model-development activities.
- Applied predictive modeling and data analysis to educational workflows, helping automate decision support and reduce manual effort in recurring processes.

PROJECTS

NutriPedia (TensorFlow, Inception-V3, GPT-3.5, GCP Storage, SQL, MERN)

- Built a multimodal nutrition application combining Inception-V3 image classification with GPT-3.5-generated dietary insights, linking computer-vision predictions, structured data, and user-facing workflows.

Telco Customer Churn Pipeline (Python, XGBoost, MLflow, GitHub Actions, AWS ECR/EC2)

- Built an end-to-end churn pipeline spanning preprocessing, XGBoost training, MLflow experiment tracking, deployment, and production monitoring.
- Improved model F1 by 15% through feature engineering; automated testing and deployment with GitHub Actions and AWS ECR/EC2, reducing dev time by 30%.

Chronic Kidney Disease Predictor (Scikit-learn, Flask, MongoDB, MERN, Python)

- Developed a chronic-kidney-disease prediction model using Scikit-learn, achieving 98% F1 and translating patient features into reliable risk predictions.
- Built an end-to-end healthcare application connecting model inference, Flask APIs, MongoDB persistence, and a MERN-based user interface for real-time screening support.

Beverage Management System (YOLOv5, PyTorch, Streamlit, PostgreSQL, Docker)

- Built a real-time YOLOv5 computer-vision system achieving 98% detection accuracy for automated beverage recognition and inventory tracking.
- Integrated PyTorch inference with Streamlit, PostgreSQL, and Docker, creating a containerized, reproducible workflow for real-time inventory operations.

TECHNICAL SKILLS

Generative AI & NLP

GPT-3.5, FLAN-T5, BERT, RoBERTa, DistilBERT, Hugging Face Transformers, Prompt Engineering

Machine Learning

Scikit-learn, XGBoost, Random Forest, Classification, Regression, Feature Engineering, Model Evaluation

Deep Learning & Vision

TensorFlow, PyTorch, CNNs, Inception-V3, YOLOv5, Computer Vision

AI Systems & MLOps

MLflow, FastAPI, Flask, Model Serving, Docker, CI/CD, MLOps, Microservices

Cloud

AWS ECS, Fargate, EC2, ECR, S3, GCP, Azure

Languages

Python, SQL, R, Java, JavaScript/TypeScript, C++, Go

Databases

PostgreSQL, MySQL, MongoDB, Redis, Redshift

EDUCATION

M.S. Statistics and Data Science, University of Central Florida (GPA: 4.00)

August 2025 - Present

B.E. Computer Science and Engineering (Data Science), Osmania University (GPA: 3.66)

July 2020 - July 2024

CERTIFICATIONS & ACHIEVEMENTS

- Advanced Certification in AI & Machine Learning - IIIT Hyderabad | April 2025 - November 2025